

Intro

Sample space & probability

Probability definition

↳ in terms of frequency of occurrence

↳ frequency interpretation is not always appropriate. (intuition / belief)

Set (ensemble)

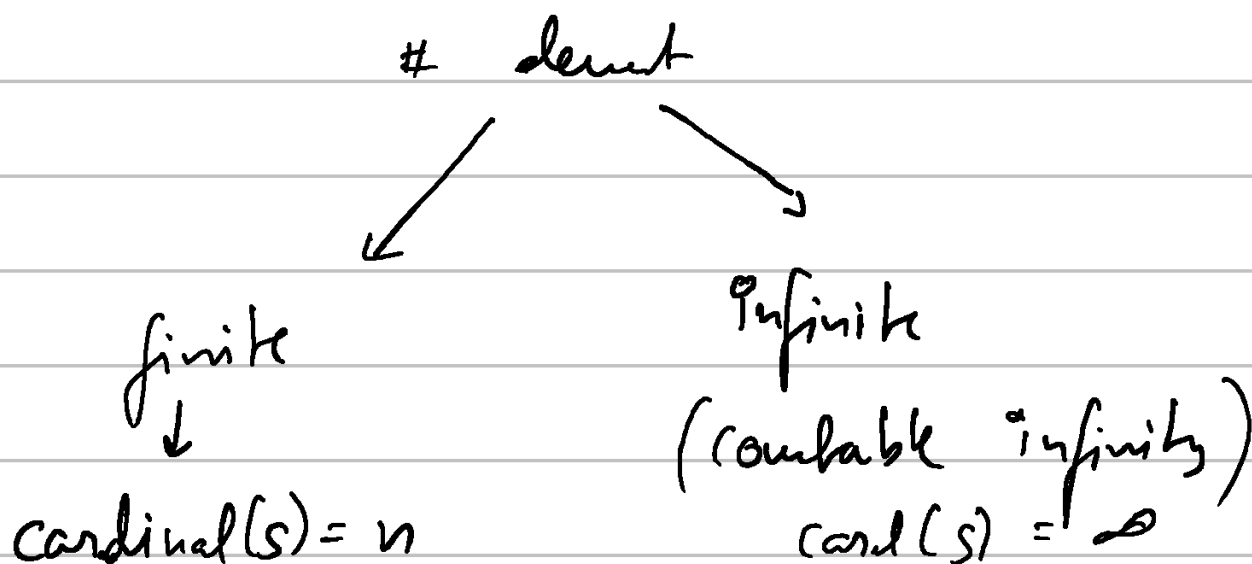
→ A collection of objects

$S \rightarrow$ set

$x \rightarrow$ element of S

$x \in S$

$\emptyset \rightarrow$ empty set.



eg $\{1, 2, 3, 4, 5, 6\}$
 $\{\text{heads}, \text{tails}\}$
 (fear) (pile)

$\text{card}(S) = \infty$

eg $\{1, 2, 3, 4, \dots\}$

We can also define a set as
 collection of all x that have a
 certain property. $\{x \mid x \text{ satisfies } P\}$
 $\hookrightarrow$ such that $\hookrightarrow$ property

$\Rightarrow$ S is contained in T

$$S \subset T$$

$$\Rightarrow S \subset T \quad \& \quad T \subset S$$

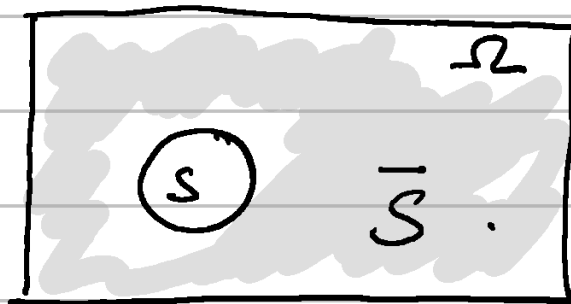
$$S = T$$

Operations

① complement

$\rightarrow$ Always with respect to something.
(universe Ω)

$$\bar{S} =$$

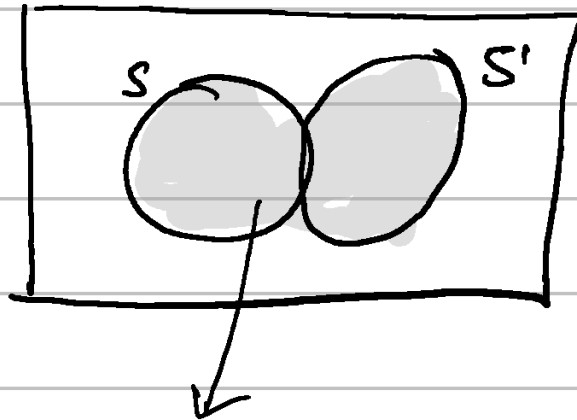


$$\overline{\Omega} = \emptyset$$

② Union

$$S \cup S'$$

↳



$$S \cup S'$$

$$S \cup T = \{x / x \in S \text{ or } x \in S'\}$$

③ Intersection

$$S \cap S' = \{x / x \in S \text{ and } x \in S'\}$$

Set algebra

- ① $S \cup T = T \cup S$
- ② $S \cap (T \cup T') = (S \cap T) \cup (S \cap T')$
- ③ $\bar{\bar{S}} = S$
- ④ $S \cup \Omega = \Omega$
- ⑤ $S \cap \phi = \phi$
- ⑥ $S \cap \bar{S} = \phi$
- ⑦ $S \cap \Omega = S$
- ⑧ $S \cup (T \cap T') = (S \cup T) \cap (S \cup T')$

De Morgan's Law

$$\textcircled{1} \quad \overline{\bigcup_n S_n} = \bigcap_n \overline{S_n}$$

$$\textcircled{2} \quad \overline{\bigcap_n S_n} = \bigcup_n \overline{S_n}$$

establishing the first law $\rightarrow$

$$\Rightarrow x \in \overline{\bigcup_n S_n}$$

$$x \notin \bigcup_n S_n$$

$$x \notin S_i \rightarrow \text{any set}$$

$$x \in \overline{S_i}$$

$$x \in \overline{\bigcap_n S_n}$$

Probability Model

A mathematical description of an uncertain situation

① sample space $\Omega \rightarrow$ set of all possible outcomes.

② Probability law which assigns to a set A of possible outcomes (an event) (a subset of universe) assigns a non-negative number $P(A)$

Axioms

① Non-negativity $P(A) \geq 0$

② Additivity if $A \cap B = \emptyset$ then
 $P(A \cup B) = P(A) + P(B)$

③ Normalisation $P(\Omega) = 1$

Discrete uniform probability law

$\Rightarrow$ All elements of discrete set are assumed to have the same probability of occurring.

$\text{card}(\Omega) = n$ equally likely.

then probability of any event A
$$P(A) = \frac{\text{card}(A)}{\text{card}(\Omega)}$$

Exercise

① Let $A, B, C \rightarrow 3$ events.
express the following events. vs A, B, C .

- Only A occurs

$$A \cap \bar{B} \cap \bar{C}$$

- At most 2 events occur among the three.

$$\overline{A \cap B \cap C}$$

② 3 dice are thrown

i) Model the experiment.

$$\Omega = \left\{ \begin{array}{l} \{1, 1, 1\}, \{1, 1, 2\} \dots \{1, 1, 6\}, \\ \{1, 2, 1\}, \dots \{1, 2, 6\}, \\ \vdots \end{array} \right.$$

$$\begin{array}{l} \{6, 6, 1\}, \{6, 6, 2\} \dots \{6, 6, 6\} \\ \rightarrow \{(i, j, k) / i, j, k \in \{1, 2, 3, 4, 5, 6\}\} \end{array}$$

$$A = \{\{1, 1, 1\}, \{2, 2, 2\} \dots \{6, 6, 6\}\}$$

$$\text{ii) } \text{card}(\Omega) = 36 \times 6$$

$$\text{card}(A) = 6$$

$$P(A) = \frac{6}{36 \times 6} = \frac{1}{36}$$

iii) Prove to have the same number on 2 dice & another on this

$$B = \{ \{1, 1, 2\} \quad \text{---} \quad \{1, 1, 6\} \} \quad \text{---} \quad \textcircled{5}$$

$$B = \left\{ (i, j, k) \mid \begin{array}{l} i=j \neq k \text{ or} \\ i=k \neq j \text{ or} \\ i \neq j = k \end{array} \right\}$$

$$\text{card}(B) = 6 \times 5 \times 3$$

$$P(B) = \frac{\text{card}(B)}{\text{card}(\Omega)} = \frac{6 \times 5 \times 3}{6^3} =$$

$$= \frac{18}{36} = \frac{6}{12}$$

iv) Probability that all 3 are different.

$$C = \{ \{1, 2, 3\}, \{1, 2, 4\}, \dots, \{1, 2, 6\} \}$$

$$C = \bar{A} \cup \bar{B}$$

$$C = \overline{A \cap B}$$

$$C = 1 - P(A) - P(B)$$

$$= 1 - \frac{1}{36} - \frac{15}{36} = \frac{20}{36}$$

$$= \frac{5}{9}$$

Properties of probab laws

$A, B, C \rightarrow \text{events}$

$$\textcircled{1} \quad \text{If } A \subseteq B \text{ then } P(A) \leq P(B)$$

$$\textcircled{2} \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\textcircled{3} \quad P(A \cup B) \leq P(A) + P(B)$$

$$\textcircled{4} \quad P(A \cup B \cup C) = P(A) + P(\bar{A} \cap B) + P(\bar{A} \cap \bar{B} \cap C)$$

Exercise

Bridge player : 52 cards

↳ Distribute 13 cards (a hand)

9) what is $\text{card}(\Omega)$

$$\Rightarrow {}^{52}C_{13}$$

$$\frac{52!}{13! (52-13)!}$$

i) Probab to have exactly one ace in my hand.

$$\Rightarrow \frac{{}^4C_1 \cdot {}^{48}C_{12}}{{}^{52}C_{13}} \rightarrow \text{cardinality.}$$

iii) $B \rightarrow$ To have at least one ace.

$$1 - \frac{{}^{48}C_{12}}{{}^{52}C_{13}} \rightarrow \text{no ace}$$

Review

i) $R \rightarrow$ At least 1 King (R_{oi})
 A $\rightarrow$ At least 1 ace (A_s)

R or A .

$$P(R \cup A) = P(R) + P(A) - \underline{P(R \cap A)}$$

$$P(R \cup A) = 1 - P(\overline{R \cup A})$$

$$= 1 - P[\bar{R} \cap \bar{A}]$$

no king and no ace

$$= 1 - \frac{{}^{44}C_{13}}{{}^{52}C_{13}}$$

v) Probab to get cards from
3 or less different color.

$$\Rightarrow P[4 \text{ different colors}] =$$

$$\frac{{}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{48}C_9}{{}^{52}C_{13}}$$

$$3 \text{ or less} = 1 - P[4 \text{ diff}]$$

$$= 1 - \frac{{}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{13}C_1 \cdot {}^{48}C_9}{{}^{52}C_{13}}$$

$$\boxed{K \cap C \cap T \cap P}$$

Exercise 4

mad postman

$n \rightarrow$ letters

$p \rightarrow$ post boxes.

$\Rightarrow x_i$ letters for box i

$$\sum x_i = n$$

Probability that box 1 is correctly filled?

$\Rightarrow \Omega \rightarrow p^n$ possibilities

$$\Omega = \{(i_1, i_2, \dots, i_n) \mid i_k \in \{1, \dots, p\}\}$$

$$\text{card}(\Omega) = p^n$$

$\Rightarrow r_1 \rightarrow {}^nC_{r_1} \rightarrow$ letters to go
in r_1

$$\Rightarrow \frac{{}^nC_{r_1} \cdot (1)^{r_1} \cdot (p-1)^{n-r_1}}{p^n}$$

$$\Rightarrow \frac{{}^nC_{r_1} \cdot (p-1)^{n-r_1}}{p^n}$$

not necessary as letters are
already marked and given
so it's just $(p-1)^{n-r}$

$$\Rightarrow \frac{(p-1)^{n-r_1}}{p^n}$$

ii) Prove that box 1 contains no other letter from other boxes.

$$0 \rightarrow (p-1)^n$$

$$1 \rightarrow {}^{r_1}C_1 (p-1)^{n-1}$$

⋮

$$i \rightarrow {}^{r_i}C_i \cdot (p-1)^{n-i}$$

$$\frac{\sum_{i=0}^n {}^{r_i}C_i (p-1)^{n-i}}{p^n}$$

⇒

$$\boxed{\frac{(p-1)^{n-r_1} \cdot p^{r_1}}{p^n}}$$

iii) Box 1 contains exactly k letters.

$$\frac{{}^n C_k \cdot (p-1)^{n-k}}{p^n}$$

iv) No empty boxes

A_i : box i is empty.

$$D : \overline{A_1} \cap \overline{A_2} \dots \overline{A_p}$$

$$D : A_1 \cup A_2 \dots \cup A_p$$

$$P(D) = 1 - P(\overline{D})$$

$$= 1 - P\left(\bigcup_{i=1}^n A_i\right)$$

$$\Rightarrow P(\overline{D}) = P(A_1) + P(A_2) \dots P(A_p) \\ - P(A_1 \cap A_2) - P(A_1 \cap A_3) \dots \\ + P(A_1 \cap A_2 \cap A_3) + \dots \\ \vdots$$

$$= \sum_{i=1}^{p-1} (-1)^{i+1} \binom{i}{p} \left(1 - \frac{i}{p}\right)^n$$

Conditional Probability

⇒ calculation of probability with some information that may affect an event.

eg 2 successive die rolls, told that sum of 2 is 9.

How likely is it that first is 6.

⇒

3	4	5	6
6	5	4	3

$\frac{1}{4}$

eg In a word guessing game,
first letter is "t". How likely
it is that the second letter
is "h"

eg how likely is it that a person
has a disease given that
a medical test was negative

⇒ Given an experiment, a corresponding
sample space & probability law.

Suppose that we know that the outcome
is within some given event B .

Quantify the likelihood that the outcome
also belongs to some other event A .

⇒ $P(A|B)$ → proba of A given B .

eg one fair die roll.

$$P[\text{outcome} = 6 \mid \text{outcome is even}] = \frac{1}{3}$$

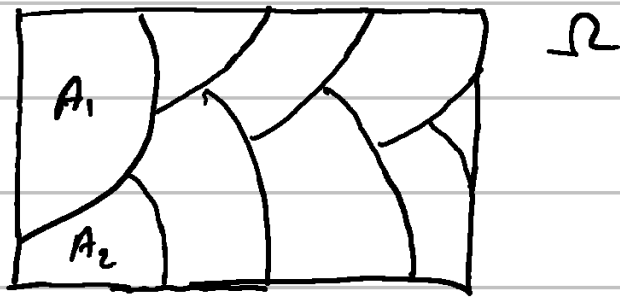
$$P(A|B) = \frac{\text{card}(A \cap B)}{\text{card}(B)}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Total probability theorem & Baye's Rule

Let $A_1, A_2, \dots, A_n$ be disjoint events that form a partition of sample space Ω

$\Rightarrow$ Partition means $\bigcup_{i=1}^n A_i = \Omega$



$$\Rightarrow P(A_i) > 0 \quad \forall i$$

$$\rightarrow P(B) = P(B \cap A_1) + P(B \cap A_2) + \dots + P(B \cap A_n)$$

$$\rightarrow P(B) = P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2) + \dots$$

$$= \sum_{i=1}^n P(B|A_i) \cdot P(A_i)$$

Bayes Rule

- Let $A_1, A_2, \dots, A_n$ disjoint events that form a partition.
- $P(A_i) > 0 \quad \forall i$

For any event B such that $P(B) > 0$

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{P(B)}$$

$$P(A_i | B) = \frac{P(A_i) \cdot P(B | A_i)}{\sum_{i=1}^n P(A_i) \cdot P(B | A_i)}$$

Independence

Conditional proba has been introduced when we have information that can affect the outcome.

What if no info is provided by the given event.

$$P(A|B) = P(A)$$

$$\begin{aligned} \& \text{ then } P(A \cap B) &= P(A|B) \cdot P(B) \\ &= P(A) \cdot P(B) \end{aligned}$$

Exercise

Cards player. A & B play cards.

B gets "r" cards from well shuffled deck.
If B has king of spade he wins.

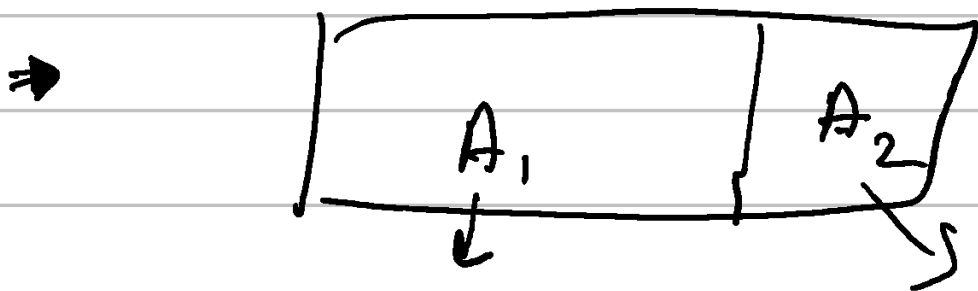
i) Prove that B wins.

$$\Omega \Rightarrow {}^{52}C_r$$

$$\text{card (win)} = 1 \cdot {}^{51}C_{r-1}$$

$$P(\text{B wins}) = \frac{{}^{51}C_{r-1}}{{}^{52}C_r}$$

ii) Player C tricks before getting r cards, he takes k cards.



spade k is
not removed

spade k is
removed

$$P(C_{\text{wins}} | A_1) = P(C | A_1) P(A_1) + P(C | A_2) \cdot P(A_2)$$

↓
0

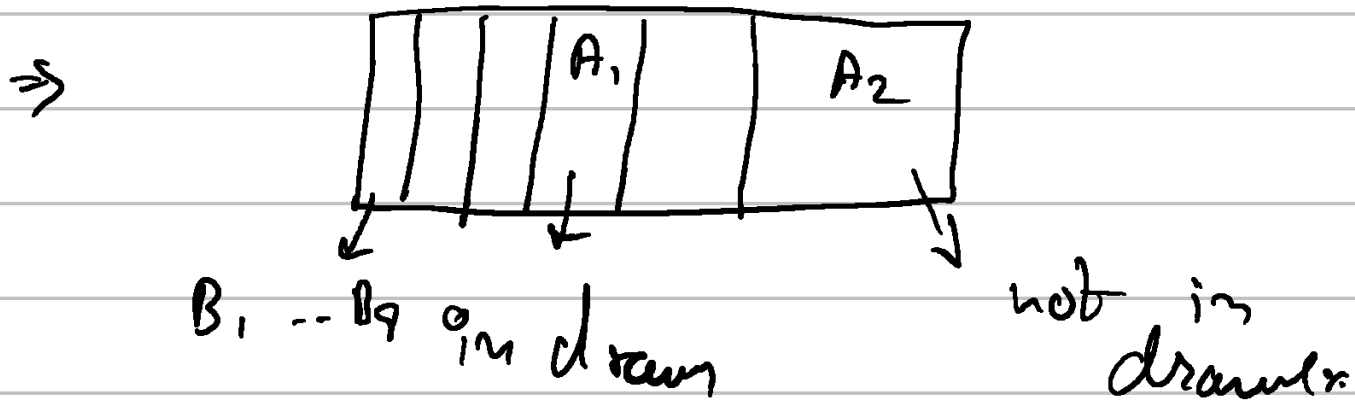
$$= \frac{{}^{51}C_{r-1}}{{}^{52}C_r} \cdot P(C | A_1)$$

$$= \frac{{}^{51}C_r}{{}^{52}C_r} \cdot \frac{{}^{51-r}C_{r-1}}{{}^{52-r}C_r}$$

Exercise

- A table has 9 drawers
- A letter has probab of $\frac{1}{2}$ to be in the table.
- If the letter is in table, each drawer has same probab to contain it.

What is proba that letter is q



$$P(B_q) = P(A_1) \cdot P(B_q | A_1) + P(A_2) \cdot P(B_q | A_2)$$

$$= \frac{1}{2} \times \frac{1}{9} = \boxed{\frac{1}{18}}$$

Another way

$$\Omega = \{D_1, D_2, \dots, D_q, \text{outside}\}$$

$$P(\text{outside}) = 1/2$$

$$P(D_1) = P(D_2) \dots = P(D_q)$$

$$P(\Omega) = P(D_1 \cup D_2 \dots D_q \cup \text{outside})$$

$$= 1$$

$$P(D_q) = \frac{1}{18}$$

ii) The first 8 drawers are opened & empty, what is probab. that letter is in 9

$$P(D_9 \mid \bar{D}_1 \cap \bar{D}_2 \dots \cap \bar{D}_8)$$

$$= \frac{P(\bar{D}_1 \cap \bar{D}_2 \dots \cap D_9)}{P(\bar{D}_1 \cap \bar{D}_2 \dots \cap \bar{D}_8)}$$

$$P(\bar{D}_1 \cap \bar{D}_2 \dots \bar{D}_8)$$

$$= P\{D_q, \text{outside}\} = 10/18$$

$$P(D_q \cap \bar{D}_1 \dots \cap \bar{D}_8) = P(D_q) = 1/18$$

$$\Rightarrow \frac{1/18}{10/18} = \frac{1}{10}$$

Exercise

In a population 7% of the people are contaminated by a virus.

We have a test having the following property.

- ① It has 99% accuracy to be positive on contaminated people.

② 3 % to be positive on healthy people.

i) What is probability that the test is positive when tested on somebody.

ii) Knowing that the test is positive what is the probability to be really contaminated.

$$\Rightarrow \text{i)} \quad P(V) = 2/100 \quad P(\bar{V}) = 98/100$$

$$P(+ve | V) = 99/100 \quad P(-ve | V) = 1/100$$

$$P(+ve | \bar{V}) = 3/100 \quad P(-ve | \bar{V}) = 97/100$$

$$P(\text{true}) = P(\text{true} | v) \cdot P(v) + P(\text{true} | \bar{v}) \cdot P(\bar{v})$$

$$= \frac{99}{100} \cdot \frac{7}{100} + \frac{3}{100} \cdot \frac{93}{100}$$

$$= \frac{972}{1000} = 0.972$$

$$\text{ii) } P(v | \text{true}) = \frac{P(v) \cdot P(\text{true} | v)}{P(\text{true})}$$

$$= \frac{\frac{7}{100} \times \frac{99}{100}}{\frac{972}{1000}}$$

$$= \underline{0.713}$$

$$\text{ii)} \quad P(V | -ve)$$

$$= \frac{P(V) \cdot P(-ve|V)}{P(-ve)}$$

$$\Rightarrow P(-ve) = P(-ve|V) \cdot P(V) + P(-ve|\bar{V}) \cdot P(\bar{V})$$

$$= \frac{1}{100} \cdot \frac{7}{100} + \frac{97}{100} \cdot \frac{93}{100}$$

$$= \frac{9028}{10000} = 0.9028$$

$$\Rightarrow \frac{\frac{7}{100} \times \frac{1}{100}}{\frac{9028}{10000}} = \frac{7}{9028}$$

$$\rightarrow 0.077 \%$$

